

DANIELSON RASOAVELONIRINA

Senior Data Engineer · Consultant (permanent contract)

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PROFILE

Senior Data Engineer with **8+ years of continuous experience** at eTech Consulting Madagascar, specialised in designing **robust data architectures** (Data Lakes, Data Warehouses, event-driven on GCP) and **high-performance ETL/ELT pipelines** (Apache Spark, Kafka, Airflow, Nifi, Talend). Recent evolution towards **technical leadership**: Backend & Cloud R&D Tech Lead, co-architect of the **Data & AI Security** department (RBAC/ABAC, GDPR, LLM guardrails). Strong proficiency in **Cloud (AWS, GCP)**, the **Big Data ecosystem (Kafka, Hadoop, Delta Lake)** and **BI platforms (Power BI, Qlik Sense, Tableau, SAS)**. **Holder of a Belgian work permit**, seeking a **Data Engineer / BI Lead role in Europe**.

TECHNICAL SKILLS

Data Engineering	Apache Spark · Kafka · Airflow · Nifi · Talend · Hadoop (HDFS) · Delta Lake · Event-driven architecture · Data Lake/Warehouse · Real-time streaming
Cloud Platforms	AWS (Lambda, EMR, RDS, S3, Terraform) · GCP (BigQuery, Dataflow, Cloud Run, Pub/Sub, Eventarc, Firestore, Vertex AI, Cloud Composer)
Business Intelligence	Power BI · Qlik Sense · Tableau · SAS (Base, Enterprise Guide, Visual Analytics) · SQL
Data & AI Security	RBAC/ABAC · GDPR · Encryption · Anonymisation · Workload Identity · Granular IAM · LLM Guardrails · Adversarial testing
Databases	PostgreSQL · MySQL · BigQuery · Firestore · DuckDB · NoSQL
Programming & Frameworks	Python (data-oriented) · SQL · Shell · JavaScript/Node.js · Java (Talend routines) · FastAPI · Flask · Pandas · Polars · Dask · Pydantic v2
Machine Learning & AI	TensorFlow · PyTorch · Keras · scikit-learn · Fine-tuning BERT/LLMs (LLaMA, Qwen) with LoRA · YOLO · Gemini function calling
DevOps & IaC	Git/GitLab · Jenkins · Cloud Build · Terraform · Docker · Artifact Registry · Secret Manager · VTOM · ADRs · CI/CD

PROFESSIONAL EXPERIENCE

eTech Consulting Madagascar

Antananarivo (remote, EU clients)

Senior Data Engineer · Salaried Consultant (permanent contract / CDI)

2018 → present

8+ years of continuous evolution: *Data Scientist (2018-2023) → Data Engineer & Quality Manager (2023-2025) → Tech Lead Backend & Data/AI Security (2026+)*. Internal engagements (R&D, security) and subcontracted assignments for major accounts: **Orange Madagascar TM, Orange Money, RAVATE, Argos Vétérinaire, Ghanty Groupe**.

- Tech Lead Backend & Cloud Infrastructure · Internal R&D (PoC)**
- April 2026 → present
- Backend / Cloud technical lead on an internal PoC of an AI-powered event-driven platform combining Speech-to-Text, translation, NER, conversational agent and business integrations (poly-repo architecture, GCP deployment).
- Architected an event-driven GCP architecture: Cloud Run + Pub/Sub + Eventarc + Firestore + BigQuery, deployed via Terraform (IaC).
 - Developed the public API in FastAPI (Pydantic v2, OpenAPI, JWT, observability): secure file ingestion via pre-signed Cloud Storage URLs.
 - Integrated AI models: orchestrated pipelines calling **Whisper/MMS (STT)**, **Gemini** (translation + NER + agent function calling), **Vertex AI**.
 - Set up DevOps / CI/CD: Cloud Build + Artifact Registry + Secret Manager, dev/staging/prod environments, monitoring with Cloud Logging + Trace.
 - Security by design: Workload Identity Federation, granular IAM, centralised secrets, GDPR compliance from inception.

- **Tech leadership:** architecture reviews, ADRs (Architecture Decision Records), mentoring the poly-repo team, Conventional Commits + protected branches.

► **Data & AI Security · Cybersecurity Department Initialisation**

January 2026 → present

- **Founding member** of the **5-person cross-functional team** driving the creation of eTech Consulting's new Cybersecurity department (referentials: **GDPR + OWASP LLM Top 10**).

SecData — Data Security:

- **Designed and deployed** uniform **RBAC/ABAC policies** across **3 environments** (dev/staging/prod) covering Data Lakes and Cloud projects (AWS/GCP).
- **Implemented** encryption (at rest / in transit), tokenisation and anonymisation protocols for sensitive data, ensuring **GDPR compliance**.
- **Continuous audit and monitoring** of 10 test pipelines via **GCP Cloud Audit Logs + Wiz**: anomalous access detection, integrity of ETL/ELT flows.

SecAI — AI Security (internal R&D PoC):

- **Designed Security Guardrails** for LLMs: prevention of prompt injection, jailbreaking and training data leakage.
- **Defined adversarial robustness testing** and **anti-PII cleansing pipelines** on training datasets.
- **Risk assessment** of third-party AI tools and definition of **secure deployment standards** for production models.
- **Flagship project:** design and industrialisation of a **uniform IAM policy** enforcing least privilege and traceability across the 3 eTech environments.

► **Data Engineer & Data Quality Manager · Orange Madagascar TM Engagement**

April 2023 → December 2025

- **Architected and industrialised 170 Apache Airflow DAGs + 184 Talend jobs** consolidating **25 multi-system sources** (GSM data, CRM, billing, network) handling **350 million rows/week**.
- **Migrated and optimised** the Data Lake / Data Warehouse architecture to **Delta Lake on Hadoop: 3× gain** on critical query latency, native ACID guarantees and time-travel.
- **Orchestrated and supervised** workflows via **Apache Airflow + VTOM**, achieving a **90 % delivery SLA** on daily business deliverables.
- **Designed and industrialised 35 decision-making dashboards** (Power BI, Qlik Sense, QlikView, Tableau) deployed to **Marketing, Finance, Risk and Executive teams** for near-real-time business monitoring.
- **Implemented an automated Data Quality framework** (45 business rules + ML-based anomaly detection): **18 blocking anomalies detected per month**, securing the reliability of indicators reported to Executive teams and to regulators (ARTEC).
- **Flagship project:** refactored the GSM data pipeline using **Spark Structured Streaming**, dividing compute time by 3 and unlocking real-time analysis on network events.
- **Mentored 6 junior engineers** on Apache Airflow and Delta Lake best practices: code reviews, DAG design, error handling, ACID patterns.

► **Data Analyst · Orange Money Madagascar Engagement (SAS)**

April 2022 → April 2023

- **Designed and automated 15 strategic dashboards** in **SAS Base, Enterprise Guide and Visual Analytics** covering Orange Money KPIs (~**180,000 transactions/day**: deposits, withdrawals, commissions, fraud), deployed to **12 users** across Marketing, Finance and Risk teams.
- **Behavioural and geographical segmentation of 725,000 subscribers** into 5 segments, foundation of retention and Marketing targeting strategies.
- **Designed and deployed a churn prediction model in production**, delivering a **35 % reduction in churn** on the population targeted by Marketing campaigns.
- **Designed and deployed a fraud scoring model in production** in close collaboration with IT, Compliance and Payment Systems teams (data governance).
- **Data preparation and cleansing in SQL + Python (pandas)** upstream of SAS analyses.
- **Delivered 5 training sessions** for 12 business users on SAS tools and results interpretation.

► **Data Manager & Backend Developer · RAVATE Engagement (Réunion)**

March 2021 → February 2022

- **Designed and deployed a hybrid AWS + on-premise cloud infrastructure** via **Terraform (IaC): 35 AWS services** across 2 accounts (dev/prod) — VPC, EC2, RDS, S3, Lambda, ALB.
- **Architected continuous synchronisation** between the **legacy on-premise ERP** and the **AWS e-commerce frontend** (50,000 product references synchronised).

- **Industrialised CI/CD: 7 AWS CodePipeline pipelines** for the data layer + **GitLab CI** for the Node.js backend + **Terraform** for infrastructure.
- **Designed and optimised a PostgreSQL (AWS RDS) database:** ~25 tables, ~10 GB, e-commerce + inventory management schema.
- **Backend development with Node.js + AWS Lambda:** ~30 REST endpoints + 10 Lambda functions for data access and manipulation logic.
- **Flagship project: full migration of the legacy infrastructure to AWS via Terraform** with a hybrid zero-downtime architecture.

► **Data Engineer · Argos Vétérinaire & Ghanty Groupe Engagements (Réunion / Montpellier)** 2020 → 2021

Two parallel engagements: Argos Vétérinaire (veterinary clinics network, France / Réunion) and Ghanty Groupe (diversified group, La Réunion).

- **Developed and maintained 35 Talend ETL/ESB jobs** for the integration and transformation of business data (orders, inventory, billing, CRM, reporting).
- **Designed 6 Apache Airflow DAGs** for orchestrating cross-application workflows.
- **Architected near-real-time synchronisation across 5 key systems** (Odoo, SFTP, legacy ERP, CRM, payment platform) supporting **~30,000 transactions/day**.
- **Flagship project 1: zero-downtime migration of the legacy ERP to Odoo** for **12 clinics** at Argos Vétérinaire.
- **Flagship project 2: real-time SFTP ↔ Odoo synchronisation architecture** with automated conflict resolution.

► **Data Scientist · eTech Internal R&D AI Projects** 2018 → 2023

Contributed to ~10 internal R&D PoC projects (3 major) in parallel with client engagements, in the fields of NLP, Computer Vision and recommendation systems.

- **Designed and trained an e-commerce product recommendation engine** (collaborative filtering + content-based) for an internal PoC.
- **Developed a topological PDF document classifier** (structure analysis + page classification) on an internal business corpus.
- **Computer Vision PoC:** object detection on **urban imagery** via **YOLO** (dataset preparation, fine-tuning, evaluation).
- **Fine-tuned BERT** for specialised entity extraction (educational NER) on a business corpus.
- **2023 extension: conversational chatbot PoC based on LLaMA-7B fine-tuned with LoRA** on consumer GPU (internal assistant), with parallel exploration of Qwen-T.

EDUCATION

École Supérieure Polytechnique d'Antananarivo (ESPA — Polytechnic School of Antananarivo) 2018 - 2019

Master 2 programme — Information Processing Systems specialisation (coursework completed)

ESPA — Polytechnic School of Antananarivo 2016 - 2017

Bachelor's degree — Telecommunications

CERTIFICATIONS

Data Engineering, Big Data & Machine Learning on GCP — Coursera (Google) 2018

BigQuery, Dataflow, Cloud Storage, Pub/Sub, Cloud Composer, Vertex AI, BigQuery ML.

Neural Networks and Deep Learning — Coursera (DeepLearning.AI) 2020

Deep Learning architectures, structuring and evaluation of ML projects.

AI Programming with Python — Udacity 2020

Data-oriented Python, Pandas, NumPy, AI fundamentals.

LANGUAGES

Malagasy: native · **French:** C2 (working language) · **English:** C1

MOBILITY & AVAILABILITY

Belgian work permit in hand · European mobility (Belgium preferred) · Onsite with relocation, hybrid or remote · Availability subject to notice period (negotiable).